

VST: A Python-based deep learning tool for segmenting electron microscopy samples

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Summary

Volume Segmentation Tool (VST) is a Python based deep learning tool designed specifically to segment three-dimensional VEM biological data without extensive requirements for cross disciplinary knowledge in deep learning. The tool is made accessible through a user-friendly interface with visualisations and a one-click installer.

Recognising the current rapid expansion of the VEM field, we have built VST with flexibility and instance segmentation in mind, hoping to ease and accelerate statistical analysis of large datasets in biological and medical research contexts. VST is composed of two main parts: the PyTorch ([Paszke et al., 2019](#))-based deep learning core that performs semantic/instance segmentation on volumetric grey scale image datasets, and a user interface that operates on top of it, responsible for constructing CLI commands to the core components for tasking. The general pipeline of VST is shown in Figure 1. We had put in efforts to ensure VST could automatically handle issues associated with large dataset sizes, instance segmentation, anisotropic voxels and imbalanced classes.

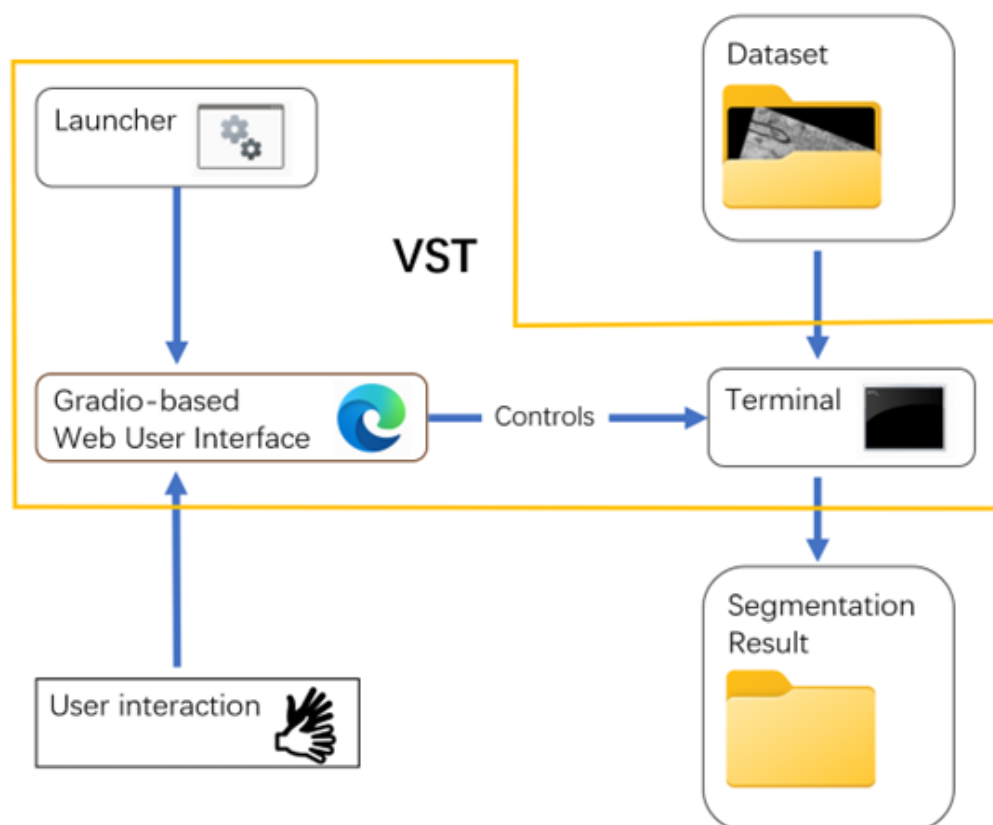


Figure 1: Schematic diagram for VST

Statement of need

Volume Electron Microscopy (VEM) enables the capture of 3D structure beyond planar samples, which is crucial for understanding biological mechanisms. With automation, improved resolution, and increased data storage capacity, VEM has led to an explosion of large three-dimensional datasets. Large datasets offer the opportunity to generate statistical data, but analysing them often requires assigning each voxel (3D pixel) to its corresponding structure, a process known as image segmentation. Manually segmenting hundreds or thousands of image slices is tedious and time-consuming. Computer-aided, especially Machine Learning (ML) based segmentation is now a routinely used method, with Trainable Weka Segmentation (Arganda-Carreras et al., 2017) and Ilastik (Berg et al., 2019) being two leading options. Emerging methods for EM image segmentation are often based on Deep Learning (DL) (Mekuř et al., 2020) because this approach has potential to outperform traditional ML in terms of accuracy and adaptivity (Erickson, 2019; Minaee et al., 2021).

State of the field

Many earlier DL tools developed are highly specific to single sample types, like in connectomics (Kamnitsas et al., 2017; W. Li et al., 2017), MRI (Milletari et al., 2016) or X-ray tomography (A. Li et al., 2022), they use a subject-optimised design at the cost of adaptability to non-target datasets. Dedicated DL segmentation tools for generalised VEM data are gradually becoming available but each have short-comings. One example, CDeep3M (Haberl et al., 2018), which uses cloud computing. Although easy to use, it was designed for anisotropic data (where the z-resolution is much lower than xy-resolution) which creates limitations when applied to

isotropic data (Gallusser et al., 2022). Another example is DeepImageJ (Gómez-de-Mariscal et al., 2021), which runs on local hardware and integrates easily with the ImageJ suite (Schneider et al., 2012). However, it only supports pre-trained models and does not have the functionality to train new ones. ZeroCostDL4Mic (Von Chamier et al., 2021), utilises premade notebooks running on Google Colab, but it requires user interaction during the entire segmentation process, which can take hours and thus is inconvenient. A more recent and advanced example is nnU-Net (Isensee et al., 2021), which auto-configurates itself based on dataset properties and has a good support for volumetric dataset, but it focuses exclusively on semantic segmentation and lacks a user friendly interface.

In short, there is a lack of tools that can handle a wide range of VEM data well for generating both semantic and instance segmentation, while at the same time being easy to use, scalable and can be run locally. This gap motivated us to develop VST - an easy-to-use and adaptive DL tools specifically optimised for generalised VEM image segmentation.

When VST was developed, BiaPy (Franco-Barranco et al., 2025) and DL4MicEverywhere (Hidalgo-Cenalmor et al., 2024) emerged as competitive options that satisfied the above criteria. They share features such as a GUI and semantic and instance segmentation on 3D images. Apart from not relying on Docker and specialising solely in 3D semantic and instance segmentation, VST incorporates more recent and sophisticated deep learning techniques to improve speed and accuracy (see the sections below).

Software design

The core principles of VST lie in the user-friendliness and scalability. The software comes with a one-click installer and full documentation on all user-accessible features, with the aim to enable accessibility for domain experts without machine learning expertise. In terms of scalability, VST uses Zarr (Miles et al., 2020), a framework for distributed storage, which allows just-in-time, chunked access for datasets much larger than the user's system memory. This is a common situation within VEM, where datasets of hundreds or thousands of gigabytes scales are present.

In terms of design, the software provides a graphical interface over a set of scripts handling various aspects of the deep learning and inference workload. The internal training framework utilises PyTorch (Paszke et al., 2019) and , the interface compiles terminal commands and activates the Python scripts as needed (Figure 1). For the DL model underneath, VST uses a 3D version of Swin transformer (Liu et al., 2021) (Wightman, 2019). This is a proven DL architecture for a wide range of vision tasks and is known for its superior performance to the traditional U-Net (Ronneberger et al., 2015) architecture, as well as its linear computational complexity. The size, depth, and other details of the model are configured automatically based on the characteristics of the user's dataset. Another recent technique that VST has integrated is the use of adaptive muon (Si et al., 2025) as the optimiser due to its faster convergence.

Much of VST's internal logic is optimised for single-class semantic and instance segmentation and cannot easily be transferred to the multi-class case or 2D image segmentation. This trade-off was made to keep workloads manageable, simplify the codebase, and maximise the ease of use for those with minimal machine learning expertise.

Research impact statement

VST has been used in postgraduate projects at the University of Otago in New Zealand for segmentation of the entire mitochondrial complement of tumorsphere (Jadav et al., 2023), as well as poorly demarked cell remnants within wool fibres. VST's competitive performance to nnU-Net (Huang et al., 2025), an MIT open-source licence, and comprehensive documentation make it ready for use by the wider community.

The graphical user interface

VST's GUI is supported by the Gradio package ([Abid et al., 2019](#)) and hosted on the user's browser.

The GUI is divided into three sections: Main, Activations Visualisation and Extras.

The main section (Figure 2) contains settings regarding training and using segmenting networks. Two segmentation modes are supported: semantic segmentation, in which the foreground objects are separated from the background, and instance segmentation, in which individual foreground objects are separated from each other as well. User can either train a new network, load an existing network and use it for predictions on new data, or train one and use it immediately.

Upon training, it automatically opens a TensorBoard interface ([Pang et al., 2020](#)) to provides various real time visualisations for the training process.

Main Activations Visualisation Extras

Segmentation Mode
☒ Semantic ☐ Instance

Components to enable
☐ Training ☐ Validation ☐ Test ☐ Predict

Workflows Explanation

Dataset Information Validation Settings Training Settings Test Settings Predict Settings

Necessary questions regarding the dataset in order to compute some hyperparameters.
 Should always be filled.

Size to spot feature
 Gives a square shaped 2d patch randomly taken from your dataset, what's the minimum side length (in pixels) you (as a human) would need to tell and segment the cellular structure of interest from the patch?
 Is used to compute the network's patch size, aka field of view.

Z-resolution to XY-resolution ratio
 The ratio of the z-resolution of the images in the dataset to their xy-resolution. We assume xy has the same resolution.

☐ Check to allow manual editing of the parameters below, instead of being calculated automatically

Example of various field of view

Automatically computed hyperparameters

☐ Enable Unsupervised Learning
 Allow using unlabelled data to enhance performance.

Calculate Training Repeats and Epochs!

☐ Read Existing Model Weight File
 Else it will create a new model with randomised weight.

Path to Existing Model Weight File, it should be a file with '.ckpt' at the end.

☐ Memory Saving Mode
 If you are experiencing running out of system memory (Not CUDA memory) during training, This option could help by using only 1 thread to do data loading. Can significantly slow down the training if your system have low single core performance.

Precision
 fp16 or bfloat16 precision could significantly cut the VRAM usage. bfloat16 is recommended over fp16 if you are using a newer GPU.

Find suitable precision based on your GPU

Model Architecture
 UNetResidual_Recommended

Base Channel Count
 Often means the number of output channels in the first encoder block. Determines the size of the network. Preferably a multiple of 8.

Use a preset formula to find the largest channel count that doesn't result in an Out-of-memory error. Isn't always accurate, only a rough estimate.

Automatically find the largest channel count

Visualising example network output on the fly
 Note: Gradio doesn't support direct display of 3D image. The result are displayed in the tensorboard.
 Could slow down training process, especially if the image is big.
 Highly recommend cropping this image into the same size as the patches that feeds into the network.

☐ Enable Visualisation

Path to the input image

File Name for Model Saved, do not include extension

Path to Save the Model Weight
 For path with space in it, put " " on both sides

Start! **Interrupt!**

Use via APT Built with Gradio Settings

Figure 2: The main interface of VST

The activations visualisation section requires a trained network and an example image. Given that image, it plots the activation across each channel through all layers of the network.

The extra section contains two functionalities: exporting the TensorBoard log to an Excel table, calculating segmentation metrics between (potentially) generated labels and ground truth labels.

AI Usage Disclosure

ChatGPT and DeepSeek were used to generate some Python functions. All generated functions were thoroughly analysed, tested with real-world data, modified and verified to satisfy desired

input and output conditions. No generative AI tools were used in the writing of this manuscript, or the preparation of supporting materials.

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